

Project Summary

Geometrical Parameters in Isogeometric Analysis and Reduced Order Modeling for Structural Dynamics

Master 2 Thesis & Internship Synthesis

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Executive Summary & Core Breakthrough

This project develops an ultra-fast, client-side simulation framework for parameterized transient structural dynamics. Resolving dynamic responses under varying geometry in real time (≥ 60 FPS) is traditionally blocked by:

1. **Geometric discretization errors** from polygonal finite element meshes.
2. **Snapshot vector space incompatibility** caused by physical domain remeshing across parameters μ .
3. **Online non-affine integration bottlenecks** during finite element assembly.

By synthesizing **Isogeometric Analysis (IGA)**, **Diffeomorphic Reference Domain Pullback Mapping $\hat{\Psi}$** , and **Energy-Conserving Sampling and Weighting (ECSW)** hyper-reduction, this framework achieves:

- **Zero CAD geometric approximation error** via exact quadratic NURBS.
- **100% parameter-invariant snapshot representation** on reference domain $\hat{\Omega}$.
- **64.5 $\times$ speedup** over the Full Order Model (2.2 ms vs. 142.1 ms assembly time; ≈ 1.30 ms total solve time per step) with relative L_2 error under **0.054%**.
- Real-time, zero-install deployment in client-side WebGL (<https://cnam-internship-mumen.netlify.app/>).

1 Problem Statement and Motivations

Real-time structural dynamics simulations are essential for digital twins, interactive educational tools, and fast design optimization. In traditional Finite Element Methods (FEM):

- **Mesh Regeneration Cost:** Modifying geometric shape parameters $\mu = (x_c, r)^T$ requires complete remeshing, generating varying node counts and destroying vector alignment across parameter instances.
- **Standard ROM Failure:** Proper Orthogonal Decomposition (POD) requires a constant snapshot matrix size $\mathbf{S} \in \mathbb{R}^{N \times M}$. Parameter-dependent mesh topologies make naive POD mathematically undefined.
- **Computational Burden:** High-fidelity 2D dynamic simulations ($N = 6,888$ DOFs, $E = 3,200$ elements) require ≈ 142.1 ms for matrix assembly and up to ≈ 450 ms per time step, far too slow for real-time interactive rendering (< 16.6 ms required for 60 FPS, < 1.5 ms solve target).

2 The Three Core Methodological Pillars

2.1 Pillar 1: Isogeometric Analysis (IGA)

Pioneered by Hughes et al. (2005), IGA adopts Computer-Aided Design (CAD) basis functions—specifically Non-Uniform Rational B-Splines (NURBS)—as the basis for both exact geometry representation and field approximation:

$$\mathbf{u}_h(\mathbf{x}, t) = \sum_{a=1}^{n_b} R_a(\mathbf{x}) \mathbf{d}_a(t), \quad R_{i,j}^{p,q}(\xi, \eta) = \frac{N_{i,p}(\xi) M_{j,q}(\eta) w_{i,j}}{\sum_k \sum_l N_{k,p}(\xi) M_{l,q}(\eta) w_{k,l}} \quad (1)$$

Key Advantages:

- Exact geometric boundaries preserved across all refinement levels.
- Higher inter-element continuity (C^{p-1} for degree p), yielding superior accuracy per degree of freedom compared to C^0 Lagrangian FEM (44% fewer DOFs and 6× faster linear solves for equivalent precision).
- Native k -refinement (order elevation before knot insertion), preserving maximum continuity across elements.

2.2 Pillar 2: Diffeomorphic Reference Configuration Mapping (Pullback)

To eliminate vector space incompatibility across varying parameter settings $\boldsymbol{\mu} = (x_c, r)^T$ (notch position and cutout depth), governing equations are pulled back from the parameterized physical domain $\Omega(\boldsymbol{\mu})$ to an invariant reference domain $\hat{\Omega}$:

$$\mathbf{x} = \hat{\mathbf{x}} + \mathbf{d}_{\text{geo}}(\boldsymbol{\xi}, \boldsymbol{\mu}) = \hat{\boldsymbol{\Psi}}(\boldsymbol{\xi}) + \mathbf{d}_{\text{geo}}(\boldsymbol{\xi}, \boldsymbol{\mu}) \quad (2)$$

Using the transformation Jacobian $\mathbf{J}_{\hat{\mathbf{x}}} = \nabla_{\hat{\mathbf{x}}} \boldsymbol{\Psi}$ and determinant $J_{\hat{\mathbf{x}}, \boldsymbol{\mu}} = \det \mathbf{J}_{\hat{\mathbf{x}}} > 0$, the continuous weak forms on $\hat{\Omega}$ become:

$$\hat{m}(\ddot{\mathbf{u}}, \delta \mathbf{u}; \boldsymbol{\mu}) = \int_{\hat{\Omega}} \rho \ddot{\mathbf{u}} \cdot \delta \mathbf{u} J_{\hat{\mathbf{x}}, \boldsymbol{\mu}} d\hat{\Omega} \quad (3)$$

$$\hat{k}(\mathbf{u}, \delta \mathbf{u}; \boldsymbol{\mu}) = \int_{\hat{\Omega}} \boldsymbol{\varepsilon}_{\hat{\mathbf{x}}}(\delta \mathbf{u})^T \mathbf{J}_{\hat{\mathbf{x}}}^{-1} \mathbf{D} \mathbf{J}_{\hat{\mathbf{x}}}^{-T} \boldsymbol{\varepsilon}_{\hat{\mathbf{x}}}(\mathbf{u}) J_{\hat{\mathbf{x}}, \boldsymbol{\mu}} d\hat{\Omega} \quad (4)$$

Impact: The spatial discretization basis functions $\hat{\mathbf{R}}$ remain strictly invariant. Parameter variations $\boldsymbol{\mu}$ only modify the local Jacobians $\mathbf{J}_{\hat{\mathbf{x}}}(\boldsymbol{\mu})$, guaranteeing that all snapshot vectors reside in $\mathbb{R}^N$ and avoiding live modal re-projections.

2.3 Pillar 3: ECSW Hyper-Reduction

Projection of the semi-discrete system onto a rank- r POD basis $\boldsymbol{\Phi} \in \mathbb{R}^{N \times r}$ ($r = 15$, capturing > 99.999% system energy):

$$\mathbf{U}(t) \approx \boldsymbol{\Phi} \mathbf{q}(t) \implies \mathbf{M}_r(\boldsymbol{\mu}) \ddot{\mathbf{q}}(t) + \mathbf{C}_r(\boldsymbol{\mu}) \dot{\mathbf{q}}(t) + \mathbf{K}_r(\boldsymbol{\mu}) \mathbf{q}(t) = \mathbf{F}_r(t) \quad (5)$$

Because the stiffness matrix $\mathbf{K}(\boldsymbol{\mu})$ is non-affine, standard Galerkin ROM still requires evaluating integrals over all $E = 3,200$ elements (154.5 ms). Energy-Conserving Sampling and Weighting (ECSW) overcomes this bottleneck by replacing full integration with a sparse, non-negatively weighted sum over an active element subset $E^* \subset E$:

$$\mathbf{K}_{r, \text{ECSW}}(\boldsymbol{\mu}) = \sum_{e \in E^*} w_e \boldsymbol{\Phi}_e^T \mathbf{K}_e(\boldsymbol{\mu}) \boldsymbol{\Phi}_e \quad (6)$$

Weights $\mathbf{w}$ and active elements E^* are computed offline via Non-Negative Least Squares (NNLS) with coercivity pre-seeding of boundary/notch elements ($\mathcal{I}_{\text{act}} \leftarrow \{e_{\text{clamp}}, e_{\text{notch}}\}$) and calibrated with a modal stiffness factor γ .

3 Quantitative Performance and Validation

A systematic benchmark on a 2D clamped-notched cantilever beam under transient sinusoidal loading yields the following comparative metrics:

Table 1: Comparative performance metrics between Full Order and Reduced Order Models.

Solver Strategy	Active Size	Rel. L_2 Error	Assembly Time	Online Speedup
Full Order Model (FOM)	6,888 DOFs / 3,200 Elem.	Baseline	142.1 ms	1.0×
Galerkin ROM (Unreduced)	15 DOFs / 3,200 Elem.	0.012%	154.5 ms	0.92×
ECSW ROM (Hyper-reduced)	15 DOFs / 28 Elem.	0.054%	2.2 ms	64.5×

Key Highlights from Results:

- **Assembly Speedup:** Hyper-reduction slashes matrix assembly time from 142.1 ms to 2.2 ms (64.5× acceleration).
- **Total Time Step Solve Time:** Reduced to ≈ 1.30 ms, well within the sub-millisecond real-time budget for smooth 60 FPS rendering.
- **Energy Conservation & Accuracy:** ECSW strictly preserves symmetry and positive definiteness, retaining high fidelity with only 0.054% relative displacement error.

4 Interactive Pedagogical Web Application

The entire online pipeline is implemented in pure client-side JavaScript / WebGL with zero server backend or plugin requirements:

- **Live Parameter Manipulation:** Real-time sliders for notch position x_c and depth r dynamically morph the geometry via reference pullback with zero remeshing latency.
- **Live Solver Switching:** Direct visual comparison between FOM, Galerkin ROM, and ECSW ROM.
- **Live Stress Contours:** Real-time Von Mises stress colormaps and tip deflection graphs at ≥ 60 FPS.
- **Live App URL:** <https://cnam-internship-mumen.netlify.app/>
- **Source Code:** https://github.com/WEHBEMUMEN/internship_report

5 Future Perspectives

Future extensions of this framework include:

1. **Geometrically Nonlinear Kinematics:** Incorporating Green-Lagrange strain tensors $\mathbf{E}$ and Second Piola-Kirchhoff stress with reduced-space Newton-Raphson solvers.
2. **3D Trivariate NURBS:** Extending 2D bivariate surface patches to complex 3D continuum structures.
3. **Multi-Patch CAD Topologies:** Coupling non-conforming patches via Nitsche’s method or Mortar element formulations.
4. **Adaptive Online Sampling:** On-the-fly basis enrichment for extreme parametric excursions.